Risha Ranjan

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Education

Carnegie Mellon University **Aug 2025 - Expected May 2027**
Robotics Institute, Candidate for Master of Science in Robotic Systems Development
Courses: Deep Learning Systems; Intro to Deep Learning; Computer Vision; Robot Autonomy; Manipulation, Estimation, and Controls; Robot Localization and Mapping

Northeastern University **Sep 2021 - May 2025**
Khoury College of Computer Science, Bachelor of Science in Computer Science and Mathematics
GPA/Honors: 3.8/4.0 | Cum Laude | Dean's Merit Scholarship | Dean's List

Technical Skills

Languages & Tools: Python, C++, CUDA, Java, MATLAB, SQL, Bash, Linux, Git, Docker
Robotics & Simulation: ROS 2, MoveIt, MuJoCo, kinematics, motion planning, SLAM, RGB-D cameras
ML & Perception: PyTorch, transformers, diffusion models, FoundationPose, SAM 3, OpenCV, NumPy

Work Experience

GE Vernova Advanced Research | Edison Program Intern - Robotics **Jun 2026 - Aug 2026**

- Built a 3D kinematic simulation of an underground digging robot using constant-curvature arc models, and designed a PD steering controller that converged to target depth within ~75 push cycles
- Proved feasibility of sonde-based heading estimation by designing a controlled rotation experiment and processing SR-24 receiver data to demonstrate reliable heading recovery across the full range

Amazon Robotics | Software Development Engineer Co-op **Jan 2024 - Aug 2024**

- Built demo software for a research-stage robotic inventory system, refactoring the backend state machine to synchronize distributed components and eliminate a 10% demo failure rate
- Ran user trials measuring task time and ease of use to evaluate usability and throughput
- Integrated research prototype software with production systems to support a real-world delivery pilot
- Presented the project to executives at an internal technical conference

Research and Projects

MS Capstone Project | Robotics Software, Project Manager **Jan 2026 - Present**

- Developed an autonomous mobile manipulation system for lab automation, integrating perception, navigation (Nav2, FAST-LIO), motion planning (MoveIt), and control into a unified ROS 2 pipeline
- Achieved 80% end-to-end success on autonomous well-plate pick-and-place (~4 min per cycle) using AprilTag-based pose estimation, hand-eye calibration, and force-torque sensor guided manipulation
- Upgrading perception to markerless 6D pose estimation by integrating SAM 3 text-prompted segmentation with FoundationPose, removing the need for fiducial markers on lab objects

Learned Trajectory Scorer for Diffusion Planners | Intro to Deep Learning **Sep 2026 - Present**

- Reproduced Diffusion Planner (ICLR 2025) baselines in nuPlan's closed-loop driving simulator as the foundation for a learned trajectory selection study
- Developing a transformer-based scorer in PyTorch that predicts safety and progress metrics for candidate trajectories, to compare against rule-based selection (PDM-Closed)

RIVeR Lab | Robotics Software Engineer Co-op **Jan 2023 - Jul 2023**

- Co-authored ISER 2023 publication on benchmarking testbeds for vehicle platooning algorithms
- Co-developed four Model Predictive Control algorithms in ROS 2 and Python for real-time longitudinal control of a vehicle convoy, validating in simulation before deploying on hardware